

TECHNOLOGY STACK ANALYSIS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Core Technology Selection

The technology stack for OptiCrop was selected to support fast execution, simple deployment, and data visualization:

Technology Component	Specific Choice	Role in Project
Opticrop Technology Stack	Python 3.10+	Data manipulation and model inference.
Programming Language	Python 3.10+	
Web Framework	Flask 3.0.0	Maintains routes, handles forms, and renders templates.
Language : Python 3.10+	Flask 3.0.0	
Language : Python 3.10+	Flask 3.0.0	
Framework : Flask 3.0.0	Flask 3.0.0	
Machine Learning	Scikit-Learn 1.3, Random Forest (classifier + regressor), K-Means	StandardScaler, RandomForest, and KMeans model p
ML : Scikit-Learn 1.3	Scikit-Learn 1.3	
ML : Scikit-Learn 1.3	Scikit-Learn 1.3	
Data AI : Pandas 2.0, NumPy 2.5	Pandas 2.0, NumPy 2.5	
Data AI : Pandas 2.0, NumPy 2.5	Pandas 2.0, NumPy 2.5	
Data AI : Pandas 2.0, NumPy 2.5	Pandas 2.0, NumPy 2.5	
Data Struct : Matplotlib 3.11, Seaborn 0.13	Matplotlib 3.11, Seaborn 0.13	Data frame cleaning, formatting, and numeric operation
Data Struct : Matplotlib 3.11, Seaborn 0.13	Matplotlib 3.11, Seaborn 0.13	
Data Struct : Matplotlib 3.11, Seaborn 0.13	Matplotlib 3.11, Seaborn 0.13	
Frontend : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	Generates performance evaluation plots.
Frontend : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	
Frontend : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	
Frontend CSS : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	Styles the dark-themed glassmorphic user dashboard.
Frontend CSS : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	
Frontend CSS : HTML5, CSS3 (glassmorphism), Bootstrap 5	HTML5, CSS3 (glassmorphism), Bootstrap 5	
Testing : pytest / Python unittest	pytest / Python unittest	
Testing : pytest / Python unittest	pytest / Python unittest	
Testing : pytest / Python unittest	pytest / Python unittest	
Version Ctrl: Git / GitHub	Git / GitHub	
Version Ctrl: Git / GitHub	Git / GitHub	
Version Ctrl: Git / GitHub	Git / GitHub	

2. Design Architecture Rationale

Flask was chosen as the web frame because it integrates easily with Python-based scikit-learn models. A local CSV and SQLite database storage layout was selected to avoid external dependencies, keeping the application lightweight and simple to run.